

MCMaster UNIVERSITY, ON, Canada

WMO: 712975

Lat: 43.2619N

Lon: 79.9204W

Elev: 374

StdP: 14.50

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.8	6.0	-10.0	3.3	1.8	-4.7	4.3	8.1	18.2	32.6	17.0	30.9	5.1	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	87.9	69.9	85.0	69.1	82.3	67.8	74.2	82.2	72.5	80.2	70.9	78.2	8.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	119.6	78.0	70.1	112.3	76.4	68.4	105.7	75.1	38.0	81.8	36.4	80.2	35.0	78.3	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.2	15.3	13.5	DB	-5.5	91.9	5.4	2.8	-9.4	94.0	-12.6	95.6	-15.7	97.2	-19.6	99.2	(4)
(5)			WB	-6.3	77.3	5.2	2.1	-10.1	78.8	-13.1	80.1	-16.1	81.3	-19.9	82.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	24.3	25.9	33.8	44.3	57.5	65.9	71.7	69.9	63.3	51.7	39.8	30.9	(6)	
	DBStd	18.55	10.75	10.64	10.10	8.12	8.56	6.58	5.34	5.05	7.09	7.89	8.25	8.36	(7)	
	HDD50	3198	797	675	508	210	23	0	0	0	0	73	319	593	(8)	
	HDD65	6649	1260	1095	966	622	267	68	7	14	117	420	756	1057	(9)	
	CDD50	2604	2	0	7	38	254	477	673	617	398	125	13	0	(10)	
	CDD65	579	0	0	0	0	33	94	215	166	65	6	0	0	(11)	
	CDH74	4225	0	0	1	17	317	722	1650	1038	441	39	0	0	(12)	
	CDH80	1112	0	0	0	1	81	198	489	222	115	6	0	0	(13)	
(14) Wind	WSAvg	6.1	6.8	7.0	6.6	7.2	6.0	5.8	5.4	5.0	5.1	6.0	6.1	6.7	(14)	
(15) Precipitation	PrecAvg	34.8	2.8	2.4	2.5	3.0	3.1	3.1	3.2	3.0	3.0	2.9	2.9	2.7	(15)	
	PrecMax	43.4	5.8	4.5	5.5	5.7	5.7	7.7	7.4	6.3	6.8	5.9	5.2	5.4	(16)	
	PrecMin	27.0	1.2	1.3	1.1	1.2	1.0	0.9	0.7	1.2	1.0	0.7	0.8	1.4	(17)	
	PrecStd	4.3	1.1	0.9	1.2	1.2	1.2	1.6	1.7	1.2	1.3	1.3	1.1	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.3	58.3	64.2	76.3	87.0	89.1	92.1	88.2	88.4	78.8	65.7	55.7	(19)	
		MCWB	53.0	50.4	53.1	58.6	67.5	69.6	72.3	71.1	72.5	64.8	55.8	51.0	(20)	
	2%	DB	48.8	50.6	58.8	69.1	82.2	85.9	88.8	85.0	83.4	72.8	60.9	50.4	(21)	
		MCWB	45.5	45.3	48.1	55.6	64.9	68.5	70.6	70.1	70.5	63.0	53.4	47.0	(22)	
	5%	DB	42.3	44.1	53.4	63.5	77.4	81.5	85.2	82.4	79.2	68.2	56.8	44.9	(23)	
		MCWB	38.9	39.8	46.9	52.3	61.2	67.1	69.1	68.8	68.1	60.1	50.5	41.5	(24)	
	10%	DB	38.0	38.9	47.6	57.5	72.2	77.9	82.2	79.9	75.1	63.5	52.7	41.5	(25)	
		MCWB	35.6	35.1	40.9	47.4	60.1	64.4	67.2	67.5	67.3	57.2	47.4	38.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.3	53.4	56.1	61.4	69.8	73.6	77.4	76.4	75.2	70.3	59.4	52.6	(27)	
		MCDB	56.2	55.7	61.8	71.4	79.7	84.2	85.3	83.5	83.8	75.8	62.8	55.8	(28)	
	2%	WB	46.0	45.8	51.5	58.3	67.0	71.1	74.5	73.5	72.8	65.2	54.3	47.3	(29)	
		MCDB	48.4	50.5	55.5	66.9	78.6	81.1	82.9	80.8	79.7	70.3	59.3	50.1	(30)	
	5%	WB	39.0	39.7	46.3	53.5	64.6	69.3	72.6	71.7	70.6	61.1	50.8	42.4	(31)	
		MCDB	41.0	43.5	53.1	61.6	73.7	77.8	80.4	78.3	76.7	65.8	55.9	45.0	(32)	
	10%	WB	35.8	35.4	41.6	48.4	61.4	67.2	70.7	70.1	68.1	58.4	47.4	38.9	(33)	
		MCDB	37.9	39.0	47.4	56.1	69.3	74.9	78.2	76.6	73.1	62.3	52.4	41.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	13.0	13.6	15.4	16.8	15.9	16.8	16.0	15.7	14.1	12.1	10.1	(35)	
		MCDBR	16.1	19.3	21.5	26.1	24.2	20.8	20.1	18.8	20.2	20.5	18.9	14.9	(36)	
	5% WB	MCWBR	15.1	16.6	15.0	16.8	11.8	9.9	9.0	8.1	9.7	12.4	13.2	13.3	(37)	
		MCWBR	15.1	18.9	19.2	24.8	20.8	19.3	17.3	15.8	17.9	17.7	17.5	15.0	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.294	0.337	0.337	0.383	0.405	0.442	0.434	0.419	0.386	0.364	0.312	0.296	(40)	
		taud	2.392	2.235	2.387	2.295	2.284	2.204	2.250	2.280	2.355	2.397	2.510	2.449	(41)	
	Ebn at Noon	Ebn at Noon	266	268	287	280	276	265	265	266	266	257	259	255	(42)	
		Edh at Noon	24	34	33	40	41	45	43	40	34	29	22	22	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	447	687	1081	1408	1715	1881	1879	1651	1319	796	496	338	(44)	
	RadStd	RadStd	45	72	116	154	174	131	135	85	123	74	67	39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page